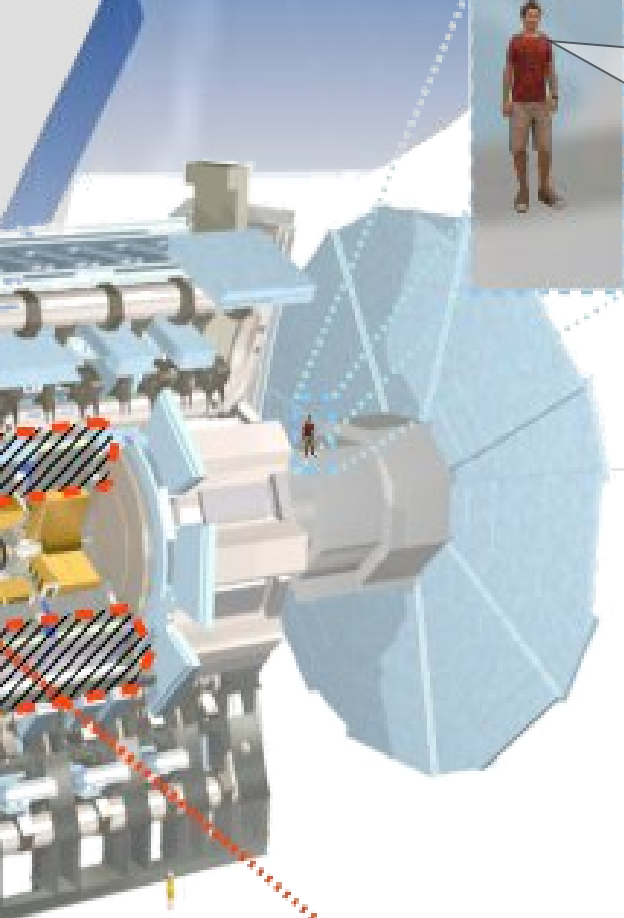




- Sampling calorimeter to identify and measure transverse missing energy, hadronic jets, hadronically decaying tau-leptons and assists in muon identification[1].
- Functionally divided in four cylindrical barrels (Fig. 1b), each with 64 modules of steel absorber plates and plastic scintillator tiles (Fig. 1c).
- Scintillator light is collected by fibers and read by electronics interfaced with PMTs positioned in the girders (Fig. 1c).
- Electronics are being upgraded to handle higher radiation and data rates by providing continuous read-out of all TileCal with better timing and energy resolution.

High Luminosity Tilecal on-detector read-out system

The on-detector electronics are segmented in 896 **Minidrawers (MDs)** that continuously sample data from all TileCal PMTs at **40 MHz** (Fig. 2). Two PMTs/cell for redundancy. Each **MD** hosts:

- **12 PMTs** powered by **High Voltage Active Dividers (HVAD)**
- **12 Front-End Boards (FEBs)** to shape and condition the PMT signals,
- a **Mainboard (MB)** to provide power received from the **Low Voltage Power Supply (LVPS)** system and continuously sample and digitize **two gains** of shaped PMT signals,
- a **Daughterboard (DB, Fig. 3)** to receive and distribute **LHC synchronized timing**, configuration and control to the front-end, and to send **continuous read-out** of the digital data from all the MB channels to the off-detector systems via multi-Gbps optical links

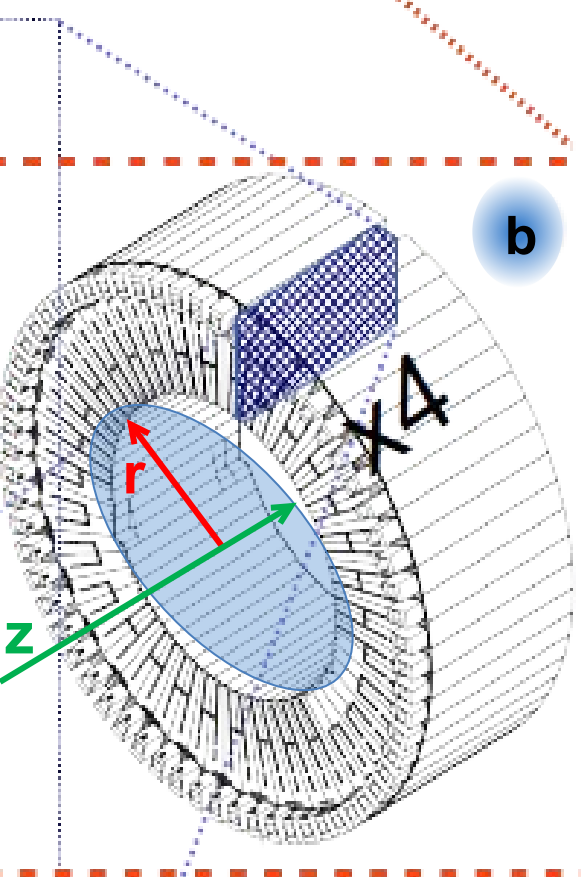
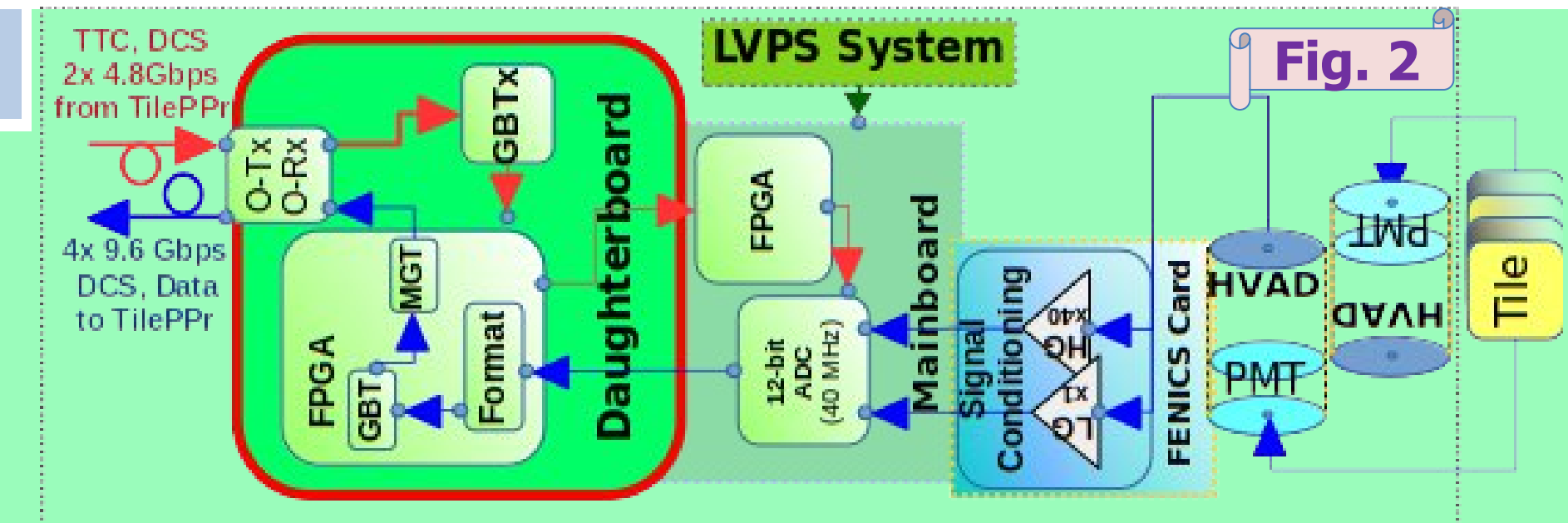
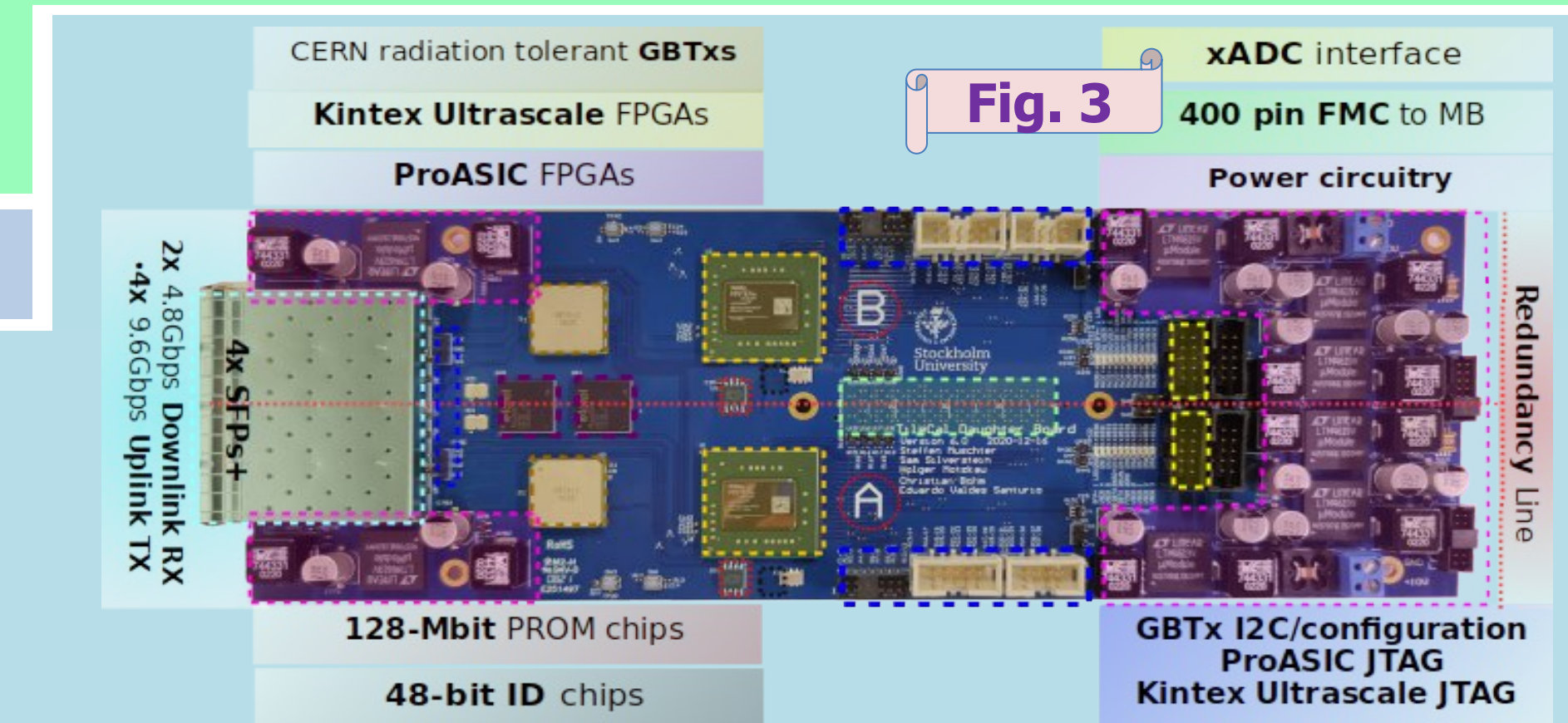


Fig. 1

The Daughterboard revision 6

- A radiation tolerant board with two functionally-equivalent halves (A and B)[2]. On each DB half:
 - An **SFP+** RX channel and a **CERN rad-hard GBTx ASIC** [3] receive a **4.8 Gbps GBT-FEC downlink** with commands and clocks from the off-detector
 - A **Xilinx Kintex Ultrascale (KU) FPGA** distributes clocks and configuration received from the GBTx to the MB, while reading slow integrator data and fast digitized data from two different gains of the PMT channels.
 - The xADCs of the KU FPGAs sample temperature, voltage stability, currents.
 - The KU FPGA sends of two copies of GBT-FEC protected words off-detector through two 9.6 Gbps uplinks via two **SFP+** TX.
 - The production of the boards will take place starting on 2026.



- Radiation tolerance was engineered into the design by adding **redundancy layers**, using the **Xilinx Soft Error Management (SEM)**, **Triple Mode Redundancy (TMR)**, **GBT-FEC** protocol, and **GBTx ASICs**.